

OMANSHI KAUSHAL

📞 +91 7060496667 ✉ E-Mail 🔗 LinkedIn 🐙 GitHub 🌐 Portfolio

AI/ML Engineer with hands-on experience building **LLM-integrated systems**, **agentic pipelines** and full-stack AI applications, spanning multimodal document intelligence, generative AI and blockchain-integrated automation.

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, SQL, Solidity, HTML/CSS

AI & ML: LLM Integration, Groq API, Claude AI, OpenAI API, LangChain, Agentic Workflows, Multimodal AI, NLP, OCR, TensorFlow, Scikit-learn, Pandas, NumPy, Feature Engineering, Classification/Regression

Frameworks & Tools: FastAPI, Next.js, React.js, Node.js, Express.js, Flask, MongoDB, Redis, REST APIs, Web3.js

Dev & Deployment: Git, Vercel, Railway, Docker, Agile, Pydantic, OpenCV, Pillow

Experience

Campus Ambassador & Developer Program Member – Accelchain 🚀

Aug 2024 – May 2025

Decentralized Systems and AI-Integrated Applications

- Centralized identity systems lacked intelligent automation, creating security gaps and slowing decentralized onboarding.
- Engineered a full-stack DApp integrating **AI-driven smart contract automation** using Solidity, Node.js, and Express.js, owning end-to-end architecture, deployment, and agile delivery pipeline.
- Achieved **40% improvement** in verification efficiency, reducing onboarding friction and scaling identity resolution across distributed workflows.

Key Projects

MemeForge – AI-Powered Meme Generation Platform | *Next.js, FastAPI, Groq API, OpenCV* 🚀 [Live Demo](#)

April 2026

- Free-form text prompts lacked any pipeline to generate contextually relevant, viral-style memes autonomously.
- Built an end-to-end AI pipeline using Groq LLM for caption generation combined with an OpenCV + Pillow image processing engine.
- Generated context-accurate memes across **10+ templates** with ~800ms average latency, fully automated from raw text prompt.

StructurAid – LLM-Powered Document Intelligence Platform | *FastAPI, OCR, Redis, Python* 🚀 [Live Demo](#)

March 2025

- Unstructured documents like invoices and contracts had no automated path to machine-readable structured outputs.
- Built a multi-stage OCR + LLM pipeline using Claude AI for schema inference and Pydantic validation across 4 document types.
- Reduced document-to-JSON conversion time by **70%** with exportable JSON/CSV outputs across all document categories.

ChatPDF – Multimodal Document Analytics & NLP System | *Python, OpenAI API, LangChain* 🚀

Dec 2025

- Extracting insights from long unstructured PDFs required manual effort with no intelligent real-time assistance.
- Designed a **context-aware, multimodal AI app** using semantic search and LLM reasoning for real-time document Q&A, analogous to Copilot's voice and screen-context intelligence.
- Improved document comprehension efficiency by **75%**, enabling query-based information retrieval at scale.

Hackathons

Amazon Sambhav Hackathon – Pre-Finalist

Nov 2024

XportSync – AI-Augmented Blockchain Cross-Border Trade Automation

- Cross-border trade lacked intelligent automation, real-time compliance inference, and transparent decentralized tracking.
- Architected a **Solidity + AI-integrated pipeline** to autonomously validate trade agreements, predict compliance risks, and enable real-time shipment tracking across distributed services.
- Reduced manual trade processing by **60%** and improved compliance accuracy across international workflows.

Google Girl Hackathon

Apr 2024

Algorithmic Problem Solving – Data Structures and Algorithms in Java

- Contest tasks demanded runtime and memory-efficient solutions under strict computational constraints.
- Engineered optimized algorithms for graph traversal, dynamic programming, and backtracking — foundations directly applicable to **ML model optimization and search-based AI**.
- Improved coding efficiency by **25%** across **40+ problem statements**.

Education

VIT Bhopal University

2022 – 2026 (expected)

B.Tech in Computer Science and Engineering (Specialisation in Health Informatics)

Certifications & Leadership

Certifications: Stanford Machine Learning (Coursera), IBM Blockchain Developer Credit

Languages: English (Fluent), Hindi (Native), German (Elementary – A2)

Leadership

Vice President – Blockchain Club, VIT Bhopal

May 2024 – Jul 2025

Co-Lead, Content Writing Team – Blockchain Club, VIT Bhopal

Sept 2023 – Apr 2024

Core Member – NSS, VIT Bhopal

Dec 2022 – Apr 2025